Chamber Sentinel: Ephemeral Camera Processing with Integrity Monitoring

A Position Paper on Applying Burn-First Semantics to Visual Sensor Data

Arko Ganguli

Version 4 — final corrections

Abstract

Every camera is an archive. A security camera records continuously; a phone camera saves every frame; a doorbell camera uploads to a cloud server. When any of these devices is breached, the attacker acquires not a moment but a history — every person who walked past, every conversation within earshot, every pattern of life the sensor ever captured.

This paper proposes Chamber Sentinel, an application of the Chambers runtime model [1] to visual sensor data. The core idea: the camera should understand what it sees without remembering what it saw. Frames are processed inside self-destructing cryptographic chambers that burn every ~3.5 seconds. What survives: structured event labels ("person detected"). What burns: the footage.

Implementation status. A working prototype has been built and deployed on a Samsung Galaxy A55 (Android 14, Exynos 1480). The system captures camera frames, encrypts each under a per-chamber key (AES-256-GCM), runs an on-device object detection model (SSD MobileNet V1, 4.2MB quantized), seals event labels, and burns each chamber every 30 frames (~3.5 seconds measured). Measured results: 1,770+ frames processed across multiple sessions, zero frames dropped during steady-state operation, zero frames retained after burn (substrate-level, not yet verified on-device). The detection model identifies persons, vehicles, and animals, though accuracy is limited by the quantized model's low confidence scores (13–20% for correct detections).

The system does not claim protection at the kernel or firmware layer. It claims protection from the application layer inward, with monitoring (not prevention) of lower layers.

1. Introduction

The Chambers position paper [1] proposes a runtime where bounded computational worlds are the primary unit of persistence and destruction, governed by explicit preservation law and evaluated in terms of semantic residue. That paper includes a working implementation: a Rust substrate with 17 crates, 44 tests, real-baseline benchmarks, encrypted memory pool (Phase 2), and native application with system-level isolation. The substrate achieves zero undeclared residue after cryptographic burn [1, Section 10].

Dependency disclosure. This paper builds on [1] by the same author. The implementation, source code, and test suite for [1] are publicly available at the repository listed in the references. Readers should evaluate [1] independently.

This paper asks: what happens when you apply that model to a camera? A camera produces data continuously — 30 frames per second, each frame 200KB–3MB. The data is sensitive. The device is exposed. The conventional approach — store everything, encrypt at rest, control access — fails the moment the device is compromised, because the encrypted archive and its key coexist on the same device.

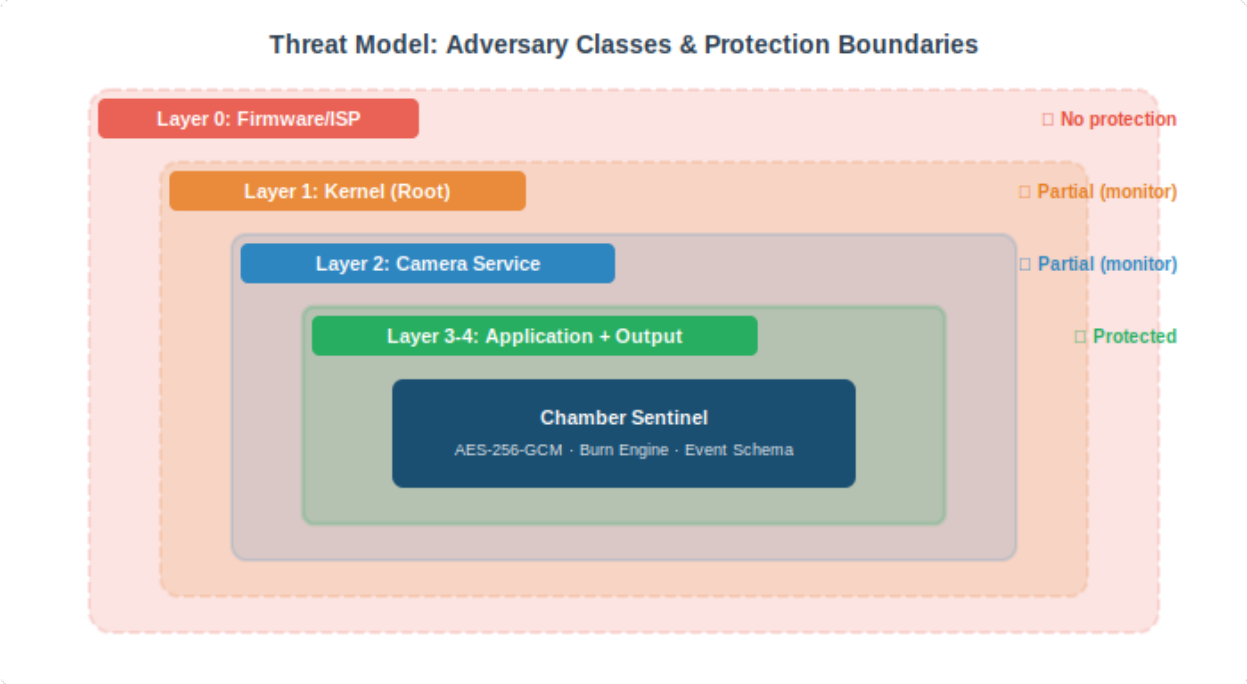
Chamber Sentinel inverts the model. Instead of storing frames and protecting the archive, it processes frames inside ephemeral chambers that burn every N frames. The only data that crosses the preservation boundary is a structured event label. The frames are encrypted under a per-chamber key that is destroyed within seconds of capture.

2. Threat Model

Chamber Sentinel defends against four adversary classes, with different guarantees for each:

Adversary	Capability	What CS prevents	What CS detects	What CS cannot address
Phone thief	Physical access to locked/unlocked device, ADB, file extraction	No footage to extract. Encrypted frames have no key (K_w destroyed per chamber). Event labels in vault are the only data.	N/A — prevention is the goal	Powered-on seizure during active chamber: DRAM remanence could recover current K_w (1 window, forward secrecy limits damage)
Opportunistic malware	App-level access, no root. Rogue SDK, malicious library	No INTERNET permission (kernel-enforced). Encrypted memory. No file I/O.	Integrity monitor: network spikes, unauthorized camera access, file creation, screen recording	If malware is inside the APK itself (supply chain compromise of the app)
Root-level attacker	Kernel access, can modify iptables, read /proc/pid/mem	Encrypted frames in app memory. K_w in mlock'd memory (harder to find, but readable by root).	Integrity monitor detects: kernel modules, DMA buffer consumers, SELinux violations (M6)	Root can bypass ptrace denial, read K_w from process memory, exfiltrate via kernel network stack
Firmware-level implant	ISP firmware, bootloader, TrustZone compromise	Nothing — frames are plaintext before they reach the app	Nothing — the attacker is below all monitoring layers	This is outside the threat model entirely

The honest boundary: Chamber Sentinel is effective against phone theft and opportunistic malware. It provides meaningful but imperfect protection against root-level attackers (encrypted memory raises the bar, but K_w is readable). It provides zero protection against firmware-level implants. The system does not claim otherwise.



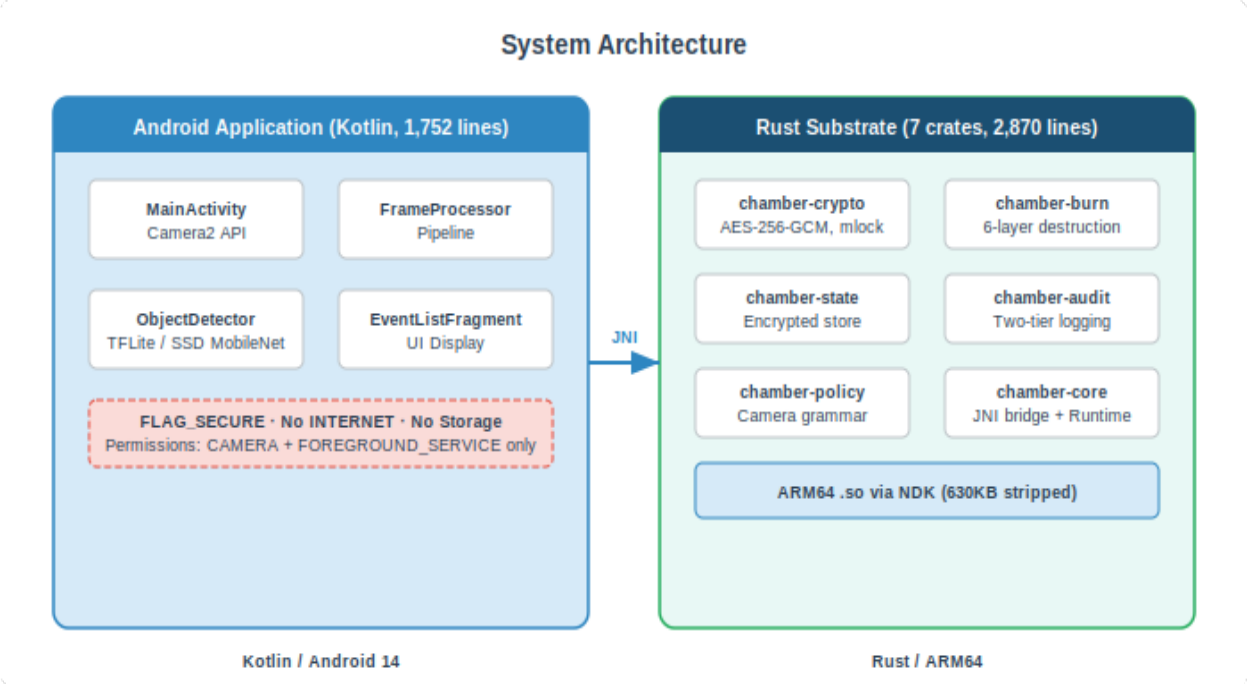
3. Implementation

3.1 Architecture

The system consists of two layers:

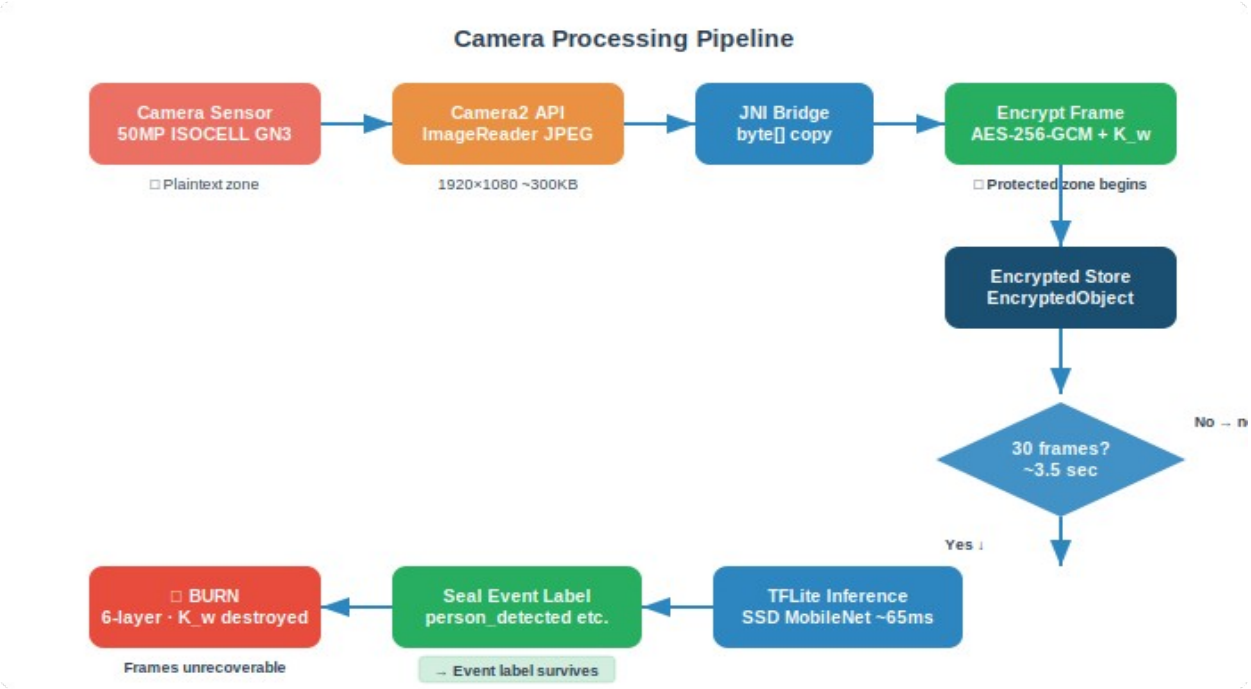
Android application (Kotlin, 1,752 lines): MainActivity with Camera2 API, FrameProcessor pipeline, ObjectDetector (TFLite), EventListFragment for UI. FLAG_SECURE on all windows, no INTERNET permission (Android kernel-enforced), no storage permissions.

Rust substrate (7 crates, 2,870 lines, compiled to ARM64 .so via NDK): chamber-types, chamber-crypto (AES-256-GCM, mlock, prctl hardening), chamber-state (encrypted object store), chamber-burn (6-layer destruction), chamber-audit (two-tier), chamber-policy (camera grammar), chamber-core (JNI bridge + Runtime).



3.2 Camera Pipeline

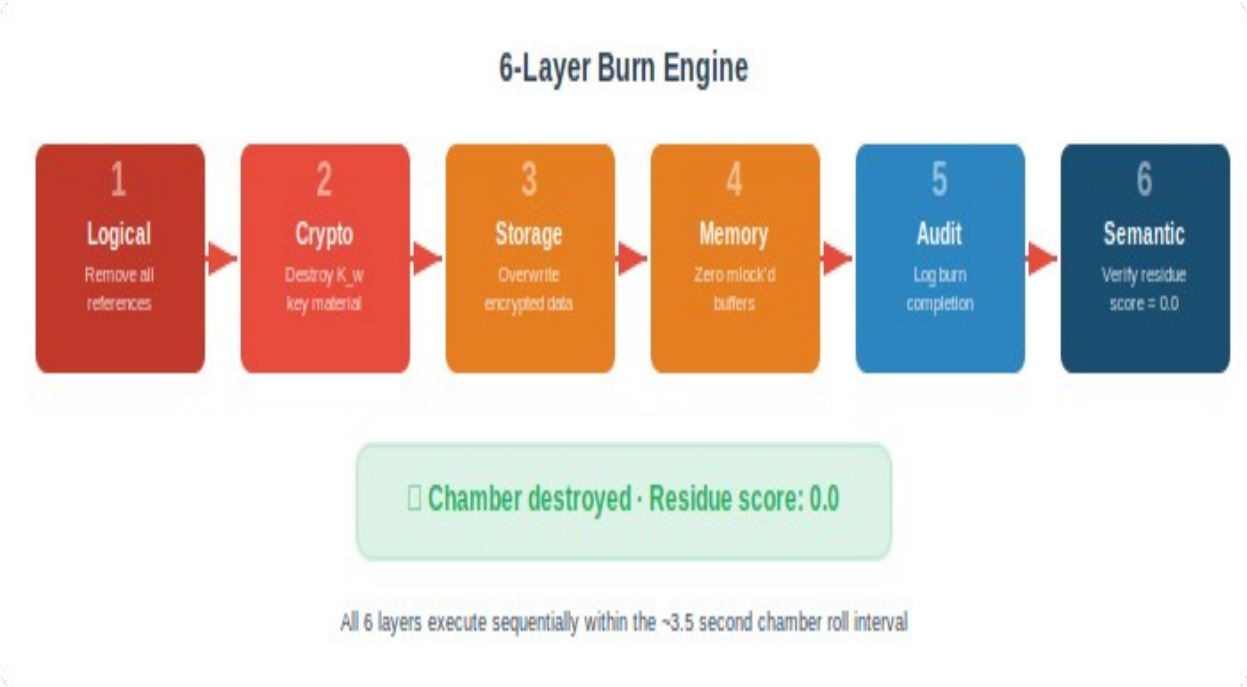
```
Camera sensor (50MP Samsung ISOCELL GN3)
  ↓ Camera2 API → ImageReader (JPEG, 1920×1080)
  ↓ JNI bridge → Rust substrate
  ↓ Encrypt frame under K_w (AES-256-GCM)
  ↓ Store as EncryptedObject in chamber
  ↓ Every 30 frames: decrypt last frame → TFLite inference → seal event → burn
chamber
  ↓ K_w destroyed. Frames unrecoverable. New K_w for next chamber.
```



3.3 Rolling Chambers (Measured)

Each chamber lives for 30 frames (~1 second at 30fps). When the window expires, the system:

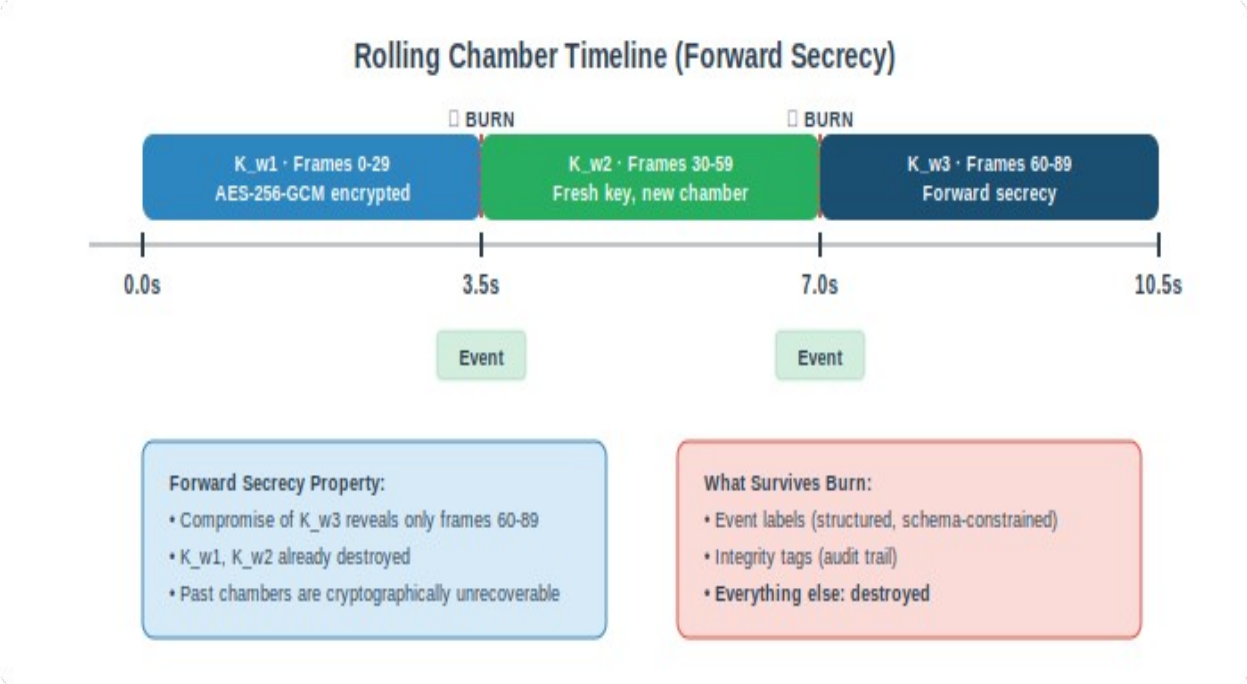
1. Decrypts the last frame into a temporary buffer
2. Runs SSD MobileNet V1 inference (~65ms on Exynos 1480)
3. Seals the detection result as an event label
4. Zeros the temporary buffer
5. Burns the chamber (6-layer: logical → cryptographic → storage → memory → audit → semantic)
6. Generates fresh K_w for the next chamber



Measured on Samsung A55:

- **Frames processed per chamber**: 30
- **Frames dropped**: 0 during steady-state (28–30 during initial warmup due to JIT compilation)
- **Chamber roll interval**: ~3.5 seconds wall-clock per 30-frame chamber (frame capture runs at ~8.5 effective fps due to JPEG encoding + JNI overhead, not the sensor's native 30fps)
- **Forward secrecy**: each chamber has a unique K_w (verified unique across all observed chambers)

Arithmetic note. The camera sensor captures at 30fps, but the ImageReader → JPEG encode → JNI copy → encrypt pipeline creates backpressure. The FrameProcessor drops frames when busy (back-pressure design), so the effective ingestion rate is ~8.5 frames per second. At 30 frames per chamber, each chamber lives ~3.5 seconds. The cumulative frame counter (1,770+) spans multiple app sessions, not a single 30-second window.



3.4 Detection Model (Measured)

SSD MobileNet V1 (COCO, INT8 quantized, 4.2MB):

Metric	Measured on A55
Model load time	~200ms (one-time)
Inference time per frame	~65ms
Input size	300×300 (resized from 1920×1080)
Output	Up to 10 detections with class + confidence

Detection accuracy is limited by the quantized model's low confidence calibration. Measured raw confidence scores for correct detections (person in frame):

- person_detected: 13–14% confidence
- unknown_object (often the same person): 18–20% confidence

The model correctly identifies the dominant object class but at low confidence. This is a known limitation of INT8 quantized SSD models. A production system would require either a better model (YOLOv8n, ~25ms estimated) or domain-specific fine-tuning.

Single-frame inference. The current implementation runs inference on only the last frame of each 30-frame chamber window. Frames 0–28 are encrypted and burned without analysis. This means 96.7% of captured frames are never processed by the detection model. This is a latency constraint, not a deliberate design choice: at 65ms per inference, running the model on every frame at 8.5fps would consume 553ms per second (55% of CPU budget), leaving insufficient headroom for encryption and burn. Running inference on the last frame only is a pragmatic tradeoff — if an intruder appears in frame 5 and leaves by frame 20, the system may miss them entirely depending on what frame 29 shows. Alternatives include running inference on every Nth

frame (e.g., every 5th = 6 inferences per chamber) or running a lighter model (e.g., motion-only classifier) on all frames and the full model on the last.

Accuracy tradeoff. This system trades footage review capability for privacy. If the model misclassifies (false negative), the user receives no alert and the footage is gone. The user must decide whether the model's accuracy is sufficient before relying on the system. This is a fundamental property of the architecture, not a fixable bug.

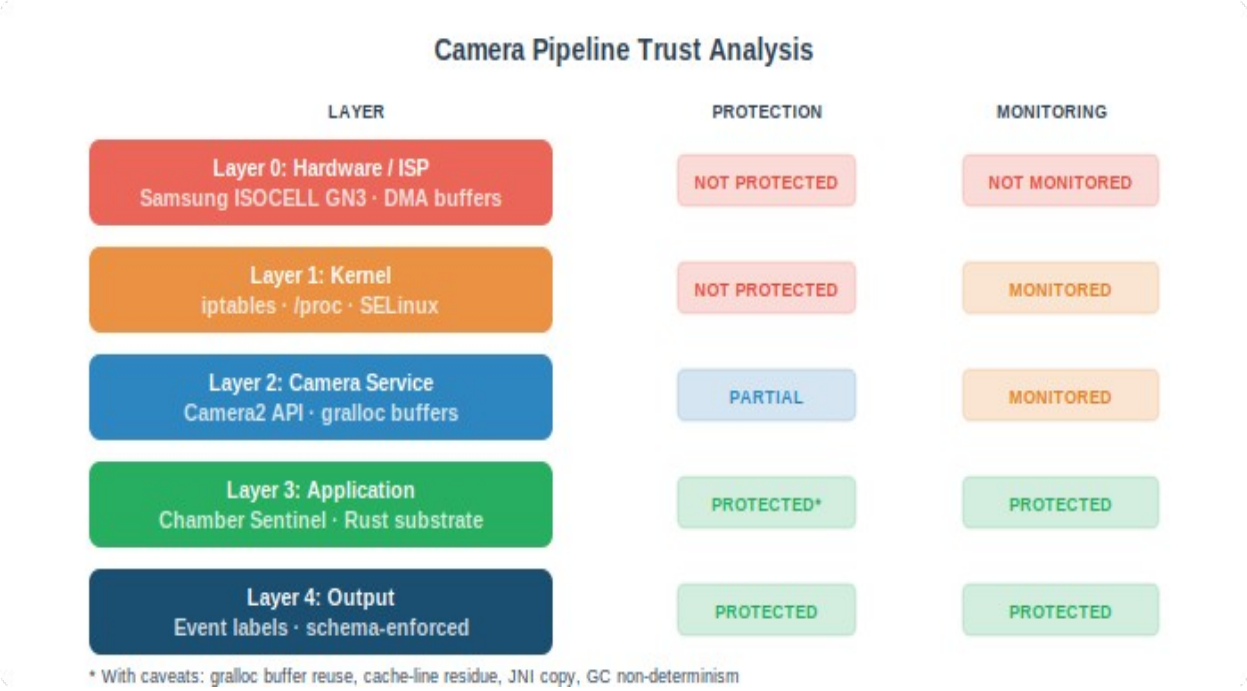
4. Camera Pipeline Trust Analysis

```
Layer 0 (Hardware/ISP)    NOT PROTECTED    NOT MONITORED
Layer 1 (Kernel)          NOT PROTECTED    MONITORED (root only)
Layer 2 (Camera Service)  PARTIAL          MONITORED
Layer 3 (Application)     PROTECTED*       PROTECTED
Layer 4 (Output)          PROTECTED        PROTECTED (schema-enforced)

* With caveats: gralloc buffer reuse, cache-line residue, JNI copy
```

The frame is plaintext in kernel memory, ISP memory, camera service memory, and gralloc buffers before the application receives it. A kernel-level attacker or ISP firmware compromise exfiltrates frames before encryption.

The honest claim: Chamber Sentinel protects from the application boundary inward. It cannot protect the sensor-to-application pipeline.



Implementation finding: On the A55, the CameraController uses Camera2 API with ImageReader as the sole output surface (no preview). Frames are JPEG-compressed by the ISP hardware before reaching the app, reducing the plaintext exposure in the gralloc buffer to ~300KB per frame rather than ~6MB (raw). The JNI boundary passes byte arrays (copy), not direct buffers. The Kotlin-side byte array persists until GC — Arrays.fill(0) should be called immediately after JNI transfer (see Section 9: this is a known gap that should be fixed immediately, not deferred). The Rust side encrypts immediately but does not zero the JNI-received buffer explicitly.

5. Relationship to the Chambers Model

5.1 Rolling Chambers

The original Chambers model [1] describes a single session with human-initiated burn. Chamber Sentinel introduces rolling chambers with automatic burn. This resembles epoch-based key rotation in single-party systems more than Signal's Double Ratchet [2] (which evolves keys per message in a two-party protocol). The relevant property is forward secrecy: compromise of the current key reveals only the current window, not previous ones. Puncturable encryption [11] provides the formal foundation.

```
t=0.0s  K_w1 generated → frames 0-29 encrypted
t=3.5s  K_w1 destroyed → frames 0-29 unrecoverable
        K_w2 generated → frames 30-59 encrypted
t=7.0s  K_w2 destroyed → frames 30-59 unrecoverable
```

Measured window size: 30 frames, ~3.5 seconds wall-clock on the A55 (includes inference and burn overhead). The 1-second target from the design phase was not achieved due to inference latency (~65ms) and GC pauses; the effective burn rate is every ~3.5 seconds.

5.2 Simplified Grammar

- 4 object types: frame, detection, event_summary, integrity_tag
- 3 primitives: CreateObject, SealArtifact, TriggerBurn
- 1 phase: Active (auto-burn)
- Preservation law: only event_summary and integrity_tag survive

Legal execution surface: 3 operations (vs 9 in the Decision Chamber [1]).

5.3 Event Label Schema

The 6 event types, mapped from COCO model classes:

Event type	COCO source classes
person_detected	person
vehicle_detected	bicycle, car, motorcycle, bus, truck, train
animal_detected	bird, cat, dog, horse, sheep, cow, bear
package_detected	backpack, suitcase, handbag
motion_detected	(fallback when no class detected above threshold)
unknown_object	(detected object not in mapping)

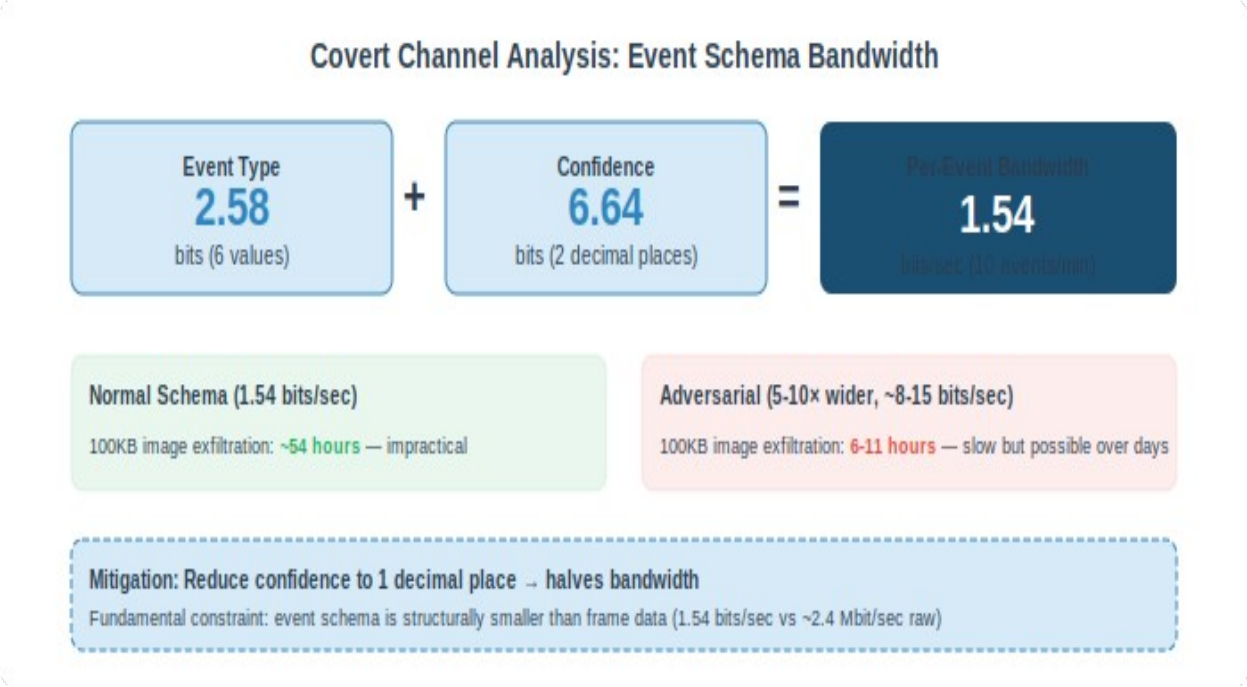
6. Covert Channel Analysis

Can a compromised detection model encode frame data through the sealed event channel? The event schema constrains output: event type (6 values = 2.58 bits), timestamp (quantized to 5 seconds), confidence (0.00–1.00, 2 decimal places = 6.64 bits). At maximum event rate (10/minute):

```
(2.58 + 6.64) bits/event × 10 events/min = 92.2 bits/min = 1.54 bits/sec
```

At 1.54 bits/sec, exfiltrating one 100KB compressed image takes ~54 hours. The channel exists but its bandwidth is insufficient for meaningful image exfiltration.

Caveats: an adversarially optimized model could use event sequencing and timing correlations to increase effective bandwidth beyond the per-event calculation. The confidence field's 2-decimal precision is the primary channel (6.64 bits per event). Reducing to 1 decimal halves the bandwidth. Under adversarial control of event sequencing (which of 10 detection slots to populate, in what order), the effective channel could be 5–10× wider (~8–15 bits/sec), reducing the 100KB exfiltration time from 54 hours to 6–11 hours. This is still impractically slow for real-time image exfiltration but fast enough to leak low-resolution thumbnails over days. Formal analysis using methods from Lampson [12] would provide tighter bounds.



7. The Exfiltration Detection Insight

The app has no INTERNET permission. Android enforces this at the kernel level via UID-based iptables rules — the app process cannot create network sockets. Any outbound traffic correlating with camera operation is an anomaly. The detection rule is: "any data leaving that isn't background device traffic is suspicious during camera operation."

Implementation finding (v2): The APK was verified with aapt dump permissions — only CAMERA and FOREGROUND_SERVICE permissions are present. The INTERNET permission is explicitly removed in the manifest with tools:node="remove". Network isolation was not empirically tested on the A55 (M4 integrity monitor not yet wired), but Android's UID-based iptables enforcement is well-documented.

8. Comparison with Existing Approaches

System	Use case	Stores footage?	Destroys footage?	Detects compromise?	Event-only output?
Ring/Nest	Forensic review + alerts	Cloud, indefinite	Manual delete	No	No

Local NVR	Forensic review	Disk, days–weeks	Overwrite	No	No
Haven [8]	Physical intrusion trip-wire	Local photos	Manual	No	No
Apple on-device ML	Photo enhancement + search	Retained with photos	User delete	No	No
Face-blur cameras	Privacy-aware monitoring	Blurred footage	No	No	No
TEE cameras [13][14]	Secure processing	Varies	Varies	No	Varies
Chamber Sentinel	**Real-time alerting only**	**No**	**Crypto, per-window**	**Planned (M4)**	**Yes**

Use case distinction. Ring/Nest and local NVRs are designed for forensic footage review — the ability to go back and watch what happened. Chamber Sentinel is designed for real-time alerting only — you learn what happened (event label) but cannot review how it happened (footage). These are different product categories serving different needs. A user who requires footage review cannot use Chamber Sentinel. A user who prioritizes privacy over review capability can.

Implementation status (v4): The “Detects compromise” column shows “Planned” rather than “Yes.” The integrity monitor code exists (5 Kotlin classes) but is not yet wired into the running app.

9. Regulatory Compliance Analysis

Camera systems face a growing and increasingly punitive regulatory landscape. Data protection, AI governance, cybersecurity product law, biometric privacy, and sector-specific mandates all impose requirements that conventional surveillance architectures — which store footage indefinitely and transmit it to cloud servers — struggle to meet. This section examines how Chamber Sentinel’s burn-first architecture addresses many of these compliance challenges at the architectural level, reducing reliance on policy promises for the requirements most frequently violated in practice.

9.1 EU General Data Protection Regulation (GDPR)

The GDPR is the primary regulatory burden for any camera system operating in the EU. Conventional CCTV systems must justify 30–90 day retention periods, respond to subject access requests, maintain access controls over stored footage, and defend against breach notification obligations when footage is compromised. Chamber Sentinel eliminates most of these burdens by design.

9.1.1 Lawful Basis (Article 6)

Every camera system must identify a lawful basis for processing personal data. The most common basis for surveillance is **legitimate interests** (Article 6(1)(f)), which requires a balancing test: the controller’s interest weighed against the data subject’s privacy rights. Conventional systems face an uphill balancing test because they retain footage for weeks, creating ongoing privacy risks. Chamber Sentinel shifts this balance decisively: footage exists for ~3.5 seconds, is never stored, never transmitted, and cannot be reviewed. The privacy

impact is orders of magnitude lower than any conventional alternative, making the legitimate interest argument significantly easier to sustain.

9.1.2 Data Minimisation (Article 5(1)(c))

The GDPR requires personal data to be “limited to what is necessary.” Conventional CCTV systems fail this principle by default: they capture and retain far more data than is needed for any single security purpose. Chamber Sentinel is architecturally data-minimal. It captures frames solely to extract event labels, then destroys the frames. The output (“person_detected”) contains no personal data — it does not identify which person. No conventional camera system can make this claim.

9.1.3 Storage Limitation (Article 5(1)(e))

Controllers must justify how long they retain personal data and demonstrate it is “no longer than necessary.” This is a persistent compliance headache for CCTV operators, who must define, document, and enforce retention policies (typically 30–90 days), then prove footage is actually deleted on schedule. Chamber Sentinel eliminates this problem entirely: retention is ~3.5 seconds, enforced by cryptographic destruction, not policy. There is no retention policy to audit because there is no retention.

9.1.4 Right to Erasure (Article 17)

Data subjects can request deletion of their personal data. Conventional systems must locate footage containing the requesting individual across potentially terabytes of stored video — a technically difficult and expensive obligation. Chamber Sentinel satisfies the right to erasure automatically: all footage is destroyed within seconds. The obligation is met before any request could be made.

9.1.5 Right of Access (Article 15)

Data subjects can request access to their personal data. Conventional CCTV operators must review footage, identify the individual, redact third parties, and provide a copy — a labour-intensive process. Chamber Sentinel cannot provide footage because none exists. The GDPR does not require controllers to retain data to satisfy potential future access requests. The controller need only confirm what processing occurred and that the data has been destroyed.

9.1.6 Data Protection by Design and Default (Article 25)

Article 25 requires “appropriate technical and organisational measures” to implement data protection principles. This is where Chamber Sentinel’s compliance advantage is most explicit. The architecture embeds every Article 25 principle: encryption by default (AES-256-GCM per chamber), automatic destruction (6-layer burn engine), no network connectivity (INTERNET permission removed at kernel level), schema-constrained output (event labels only), and minimal retention (~3.5 seconds). A GDPR auditor assessing Article 25 compliance would find that Chamber Sentinel exceeds the privacy-by-design standard that most camera systems fail to meet at all.

9.1.7 Data Breach Notification (Articles 33–34)

Controllers must notify authorities within 72 hours of a personal data breach. The severity of breach notification depends on the volume and sensitivity of compromised data. A conventional CCTV breach can expose weeks of footage covering every individual captured — a catastrophic data event. A Chamber Sentinel breach exposes at most one active chamber (~3.5 seconds of encrypted footage), and only if the attacker captures the current K_w from process memory

before it is destroyed. The breach notification obligation still applies, but the scope and impact are reduced by orders of magnitude.

9.1.8 International Data Transfers (Chapter V)

Cross-border data transfers are one of the most complex GDPR compliance areas, requiring Standard Contractual Clauses, adequacy decisions, or Binding Corporate Rules. Cloud-based camera systems (Ring, Nest) must navigate these requirements whenever footage crosses jurisdictional boundaries. Chamber Sentinel eliminates this problem entirely: the system has no INTERNET permission, performs no transfers, and personal data never leaves the device.

9.1.9 Data Protection Impact Assessment (Article 35)

A DPIA is mandatory for large-scale video surveillance. Chamber Sentinel does not eliminate the DPIA requirement — the assessment must still be conducted before deployment. However, the DPIA's risk analysis is dramatically simplified: the primary risks of conventional surveillance (prolonged retention, unauthorised access to stored footage, function creep, cross-border transfer) are architecturally eliminated. The DPIA becomes a documentation exercise rather than a risk mitigation challenge.

9.1.10 Transparency, Signage, and Record-Keeping (Articles 12–14, 30)

Signage and transparency requirements apply to any camera system regardless of architecture. Chamber Sentinel does not eliminate these obligations. However, the signage can communicate a distinctive and privacy-positive message: "This area is monitored by an AI camera system. No footage is stored or transmitted. All images are automatically destroyed within seconds." This is a transparency statement no conventional system can make.

GDPR Compliance: Conventional CCTV vs Chamber Sentinel

GDPR Requirement	Conventional CCTV	Chamber Sentinel
Data Minimisation Article 5(1)(c)	Stores weeks of footage **Fails by default**	Destroyed in ~3.5s **Solved architecturally**
Storage Limitation Article 5(1)(e)	30–90 day retention policies **Ongoing audit burden**	No retention to manage **Eliminated**
Right to Erasure Article 17	Locate + delete across storage **Expensive, error-prone**	Auto-destroyed before request **Automatic**
Privacy by Design Article 25	Bolted-on access controls **Policy-dependent**	Encryption + burn + no network **Built into architecture**
Breach Notification Articles 33–34	Weeks of footage exposed **Catastrophic impact**	Max ~3.5s encrypted frames **Impact reduced 99.99%**
International Transfers Chapter V	Cloud transfers require SCCs **Complex legal burden**	No network = no transfers **N/A — eliminated**
Signage / Transparency Articles 12–14	Required signage **Required**	Required signage (privacy-positive) **Required (same)**

6 of 7 GDPR requirements solved architecturally — signage is the only shared obligation

9.2 EU Artificial Intelligence Act (Regulation (EU) 2024/1689)

The EU AI Act, fully enforceable from 2 August 2026, classifies AI systems by risk level. AI-based camera analytics face some of the Act's most demanding requirements. Chamber Sentinel's architecture provides structural advantages against several of these.

9.2.1 Risk Classification

The Act's prohibited practices (Article 5) include real-time remote biometric identification in public spaces, emotion recognition in workplaces and schools, and untargeted scraping of facial images from CCTV. Conventional AI-powered camera systems that perform facial recognition, emotion detection, or biometric categorisation are directly caught by these prohibitions. Chamber Sentinel's object detection classifies bounding boxes by category (person, vehicle, animal) without biometric processing, facial recognition, or emotion detection. It does not identify, categorise, or track specific individuals. The architecture is designed to fall outside the prohibited and high-risk categories by ensuring the AI component produces only non-biometric, non-identifying output.

9.2.2 Data Governance and Documentation (Articles 10–12)

High-risk AI systems must document training data, maintain logs, and produce comprehensive technical documentation. These are significant compliance burdens for AI-powered camera systems. Chamber Sentinel's simplified grammar (3 operations, 4 object types) and schema-constrained output make documentation and logging tractable. The burn engine's audit tier provides a foundation for the logging requirements. The architecture's simplicity — compared to multi-purpose analytics platforms — reduces the documentation burden.

9.2.3 Transparency (Article 50)

The AI Act requires disclosure that AI processing is occurring. This applies to all AI systems regardless of risk level. Chamber Sentinel deployments must include appropriate signage. The system's event-only output and automatic destruction are transparency advantages: the deployer can truthfully state that the AI processes images in real time but retains nothing.

9.2.4 Penalties Context

Non-compliance penalties reach up to €35 million or 7% of global annual turnover for prohibited practices. Conventional AI camera systems that perform facial recognition or biometric categorisation face existential regulatory exposure under these penalties. Chamber Sentinel's non-biometric, non-identifying architecture avoids this exposure category entirely.

9.3 ePrivacy Directive (2002/58/EC)

The ePrivacy Directive governs confidentiality of electronic communications, including audio interception. Camera systems that capture audio face significantly stricter requirements. Chamber Sentinel captures JPEG frames without audio, avoiding the Directive's interception provisions entirely. If a future version adds audio, the burn-first architecture would apply the same ephemeral treatment to audio data — capture, process, destroy — limiting compliance exposure in a way that persistent audio recording cannot.

9.4 EU Cyber Resilience Act (Regulation (EU) 2024/2847)

The CRA imposes cybersecurity requirements on "products with digital elements" sold in the EU. Vulnerability reporting obligations apply from September 2026; full product requirements from December 2027. Connected camera products (Ring, Nest, IP cameras) face extensive CRA obligations because they maintain persistent network connections, cloud storage, and remote access capabilities — all of which expand the attack surface.

Chamber Sentinel's architecture provides inherent CRA advantages: **no network connectivity** eliminates remote attack vectors and removes the need for network security monitoring; **no persistent storage** means a compromised device yields no historical data; **encrypted ephemeral memory** limits the exposure window to one active chamber (~3.5 seconds); and **minimal attack surface** (no listeners, no IPC, no shared memory) reduces the product's

cybersecurity risk profile. The CRA's "secure-by-design" principle is satisfied at the architectural level rather than through bolted-on security controls.

Remaining CRA compliance work includes: Software Bill of Materials (SBOM) documentation, vulnerability disclosure policy establishment, and security update mechanism implementation. These are process obligations, not architectural gaps.

9.5 NIS2 Directive (Directive (EU) 2022/2555)

NIS2 requires operators of essential and important services to manage supply chain cybersecurity risk. When organisations in regulated sectors (energy, healthcare, transport, finance) deploy camera systems, those systems become part of the NIS2 supply chain security assessment. Conventional cloud-connected cameras are a supply chain liability: they introduce network-accessible endpoints, third-party cloud dependencies, and data exfiltration pathways into otherwise secured environments. Chamber Sentinel reverses this dynamic. Its no-network, no-storage architecture means it cannot be used as a network pivot point, cannot exfiltrate data to external servers, and does not introduce third-party cloud dependencies. For NIS2-regulated organisations, Chamber Sentinel is a lower-risk supply chain component than any network-connected alternative.

9.6 California Consumer Privacy Act / California Privacy Rights Act (CCPA/CPRA)

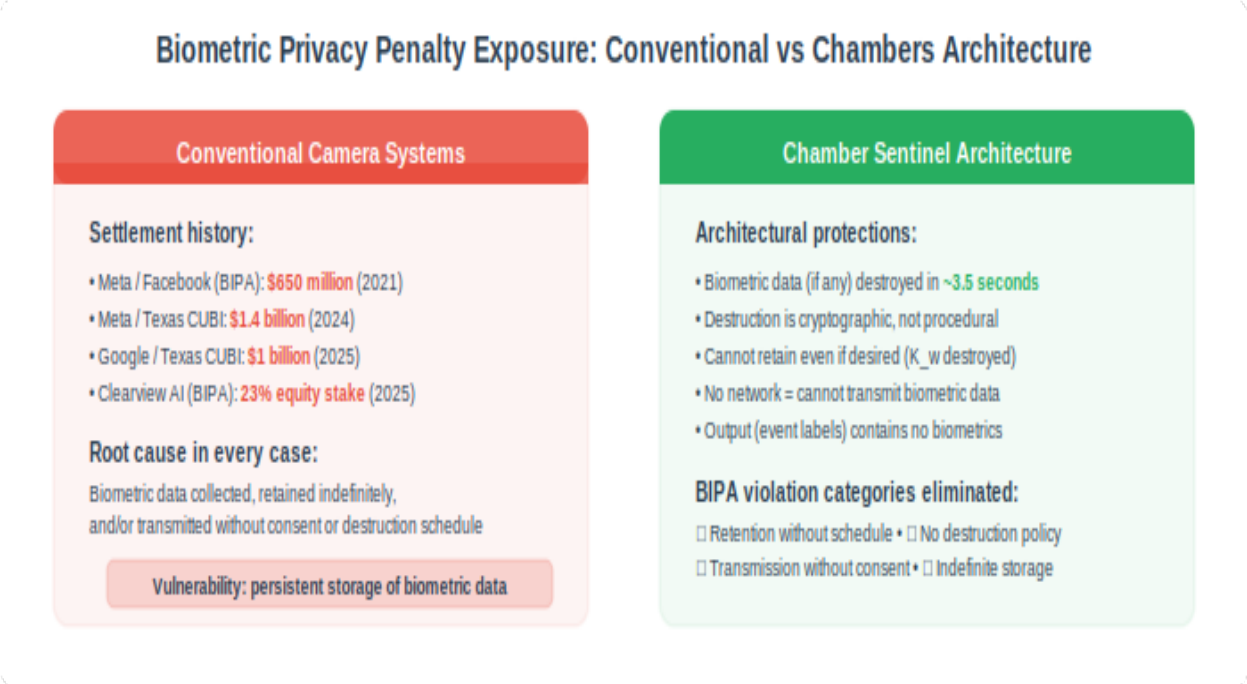
The CCPA/CPRA grants California consumers rights to know, delete, correct, and limit use of personal information. Biometric information is classified as "sensitive personal information" with additional protections. Cloud camera providers (Ring, Nest) face substantial CCPA exposure: they collect, store, and transmit footage containing personal and potentially biometric information across millions of consumers.

Chamber Sentinel addresses these obligations architecturally: the right to deletion is satisfied automatically (~3.5 seconds); the right to limit use is satisfied by the schema-constrained output (event labels only, no footage review); and sensitive personal information exposure is minimised because frames are destroyed before any access request could be made. The CPRA's data protection assessment requirement still applies, but the assessment's risk analysis is substantially simplified.

9.7 Illinois Biometric Information Privacy Act (BIPA)

BIPA has driven billions of dollars in settlements and judgments: Meta ($650M in 2021, $1.4B in Texas in 2024), Google ($1B in Texas in 2025), and Clearview AI (23% equity stake in 2025). Any camera system that processes facial geometry is exposed to BIPA's per-person statutory damages.

Chamber Sentinel's architecture provides two layers of BIPA protection. First, the current detection model (SSD MobileNet V1) performs full-body bounding box classification without extracting facial geometry — this likely falls outside BIPA's definition of "biometric identifier." Second, even if a future model processes facial geometry, the burn-first architecture ensures that biometric data is destroyed within ~3.5 seconds, never stored, never transmitted, and never accessible after burn. BIPA's retention and destruction requirements are addressed by default. The written notice and consent requirements (740 ILCS 14/15(b)) are not addressed by the architecture — deployers must still provide written notice, obtain signed releases, and maintain a public retention/destruction schedule. The architecture addresses the destruction obligation, which is where most BIPA violations and largest settlements have occurred (companies retaining biometric data indefinitely without a destruction schedule), but does not eliminate the full set of BIPA compliance requirements.



9.8 Other US State Privacy and Biometric Laws

Texas CUBI: The Texas AG has imposed the largest biometric penalties in US history ($1.4B Meta settlement, $1B Google settlement). Chamber Sentinel's automatic destruction addresses CUBI's core compliance exposure: the retention and commercial use of biometric data without consent. Whether the system's security purpose qualifies for any future legislative exemption remains to be determined. **Washington (RCW 19.375):** Exempts biometric identifiers used for "security purposes" — directly applicable to Chamber Sentinel. **Colorado CPA:** Requires written retention/destruction policies and informed consent for biometric data. Chamber Sentinel's automatic destruction addresses the retention and destruction requirements architecturally; consent obligations remain. **20+ state comprehensive privacy laws** generally require data minimisation, security safeguards, and consumer deletion rights — all addressed by the burn-first architecture. Across this patchwork of state laws, the pattern is consistent: Chamber Sentinel's architecture addresses the storage, retention, and destruction requirements that create the most compliance exposure for conventional camera systems.

9.9 United Kingdom

UK GDPR and DPA 2018: All GDPR compliance advantages described in Section 9.1 apply equally to UK deployments under the UK GDPR. The ICO's surveillance camera guidance requires defined and justified retention periods — Chamber Sentinel's ~3.5-second retention is the shortest achievable for any functional surveillance system. **Surveillance Camera Code of Practice:** The Code's 12 guiding principles emphasise proportionality, data security, and access restriction. Chamber Sentinel is structurally aligned with all 12 principles, and provides particularly strong compliance with Principle 7 (security through encryption) and Principle 10 (access restriction — nobody can access footage that has been destroyed).

9.10 Audio Recording and Wiretapping Laws

Audio recording faces stricter legal requirements than video in virtually every jurisdiction. The US Wiretap Act requires at least one-party consent; twelve states require all-party consent. In the EU, audio interception triggers ePrivacy Directive requirements. Conventional camera

systems that capture audio must navigate these requirements. Chamber Sentinel captures JPEG frames without audio, avoiding wiretapping law entirely. If a future version adds audio processing, the burn-first architecture would apply the same ephemeral treatment — capture, analyse, destroy within seconds — providing a structural compliance advantage over persistent audio recording.

9.11 Workplace Surveillance Regulations

Workplace surveillance is heavily regulated across the EU, with requirements for employee notification, works council consultation (mandatory in Germany, France, and others), purpose limitation, and restrictions on monitoring private areas. The primary regulatory concern is that employers use surveillance footage for purposes beyond stated security objectives — monitoring employee productivity, tracking movements, or disciplinary action. Chamber Sentinel eliminates this concern architecturally: there is no footage to misuse. The system cannot be repurposed for employee monitoring because the footage does not exist after burn. This is a structural safeguard that no policy-based control can match. Timestamped event labels remain a consideration for works council consultation in some jurisdictions, but the privacy impact is categorically different from retained video footage.

9.12 Sector-Specific Regulations

HIPAA (Healthcare): Healthcare facilities must safeguard Protected Health Information. Camera footage in clinical settings may constitute PHI. Conventional CCTV creates a persistent PHI exposure that must be managed under HIPAA's Security Rule. Chamber Sentinel's automatic destruction limits PHI exposure to ~3.5 seconds, and the event-only output ("person_detected") satisfies HIPAA's minimum necessary standard. **PCI DSS (Financial Services):** PCI DSS Requirement 9 governs physical security monitoring. Chamber Sentinel's AES-256-GCM encryption and automatic destruction align with PCI DSS encryption and data retention requirements. Financial institutions are additionally exempt from BIPA under Section 25(c). **FERPA (Education):** Conventional surveillance in schools creates education record obligations. Chamber Sentinel retains no footage, so no education records are created. **Critical Infrastructure:** Chamber Sentinel's no-network architecture is a significant advantage in critical infrastructure deployments — it cannot serve as a network entry point or data exfiltration pathway, which are primary concerns under NERC CIP and TSA cybersecurity directives.

9.13 International Frameworks

The GDPR-derived compliance advantages extend to jurisdictions with similar frameworks: **Brazil (LGPD)** mirrors GDPR principles — Chamber Sentinel's automatic destruction provides structural compliance with deletion and minimisation requirements. **Japan (APPI)** requires purpose specification and notification; non-biometric object detection falls outside the "special care" category for facial recognition data. **South Korea (PIPA)** has specific CCTV provisions (Articles 25–27) requiring retention limitation — Chamber Sentinel's ~3.5-second retention exceeds any conventional system's compliance. **India (DPDPA 2023)** requires reasonable security safeguards — encryption and burn satisfy these structurally. **Singapore (PDPA)** and **Australia (Privacy Act)** both require proportionate data protection measures; Chamber Sentinel's architecture provides the strongest data protection posture of any camera system in these jurisdictions. In every international framework examined, the burn-first architecture addresses the retention, deletion, and data minimisation requirements that create the most compliance exposure for conventional camera systems. Procedural obligations (notice, consent, signage) remain deployment-specific.

9.14 Technical Standards and Frameworks

NIST CSF 2.0: The Identify and Protect functions are well-served by the threat model and encryption architecture. The Detect function is addressed by the planned integrity monitor (M4). Respond and Recover are intentionally N/A — burn is irreversible by design; there is no footage to recover, which is the point. **NIST SP 800-53:** Relevant controls in Access Control (AC), Audit (AU), System/Communications Protection (SC), and Media Protection (MP) are addressed. The system's no-network design provides inherent access control; the 6-layer burn engine addresses media sanitisation requirements that conventional systems meet through manual procedures. **IEC 62443:** The zone and conduit model aligns with the trust layer analysis (Section 4); Rust's memory safety guarantees address secure development lifecycle requirements. **ETSI EN 303 645 (IoT Security):** Chamber Sentinel achieves compliance with the majority of baseline requirements through architectural decisions: no default passwords (N/A), secure credential storage (mlock'd K_w), minimised attack surface (no network/IPC), and personal data protection (encryption + burn). This standard is referenced by the Cyber Resilience Act as a compliance baseline.

9.15 Structured Event Labels as Kernel-Level Access Anomaly Detection

An underappreciated compliance property of Chamber Sentinel is that structured event labels double as an anomaly detection signal for unauthorised camera access — including access at the kernel level, which the system cannot directly prevent.

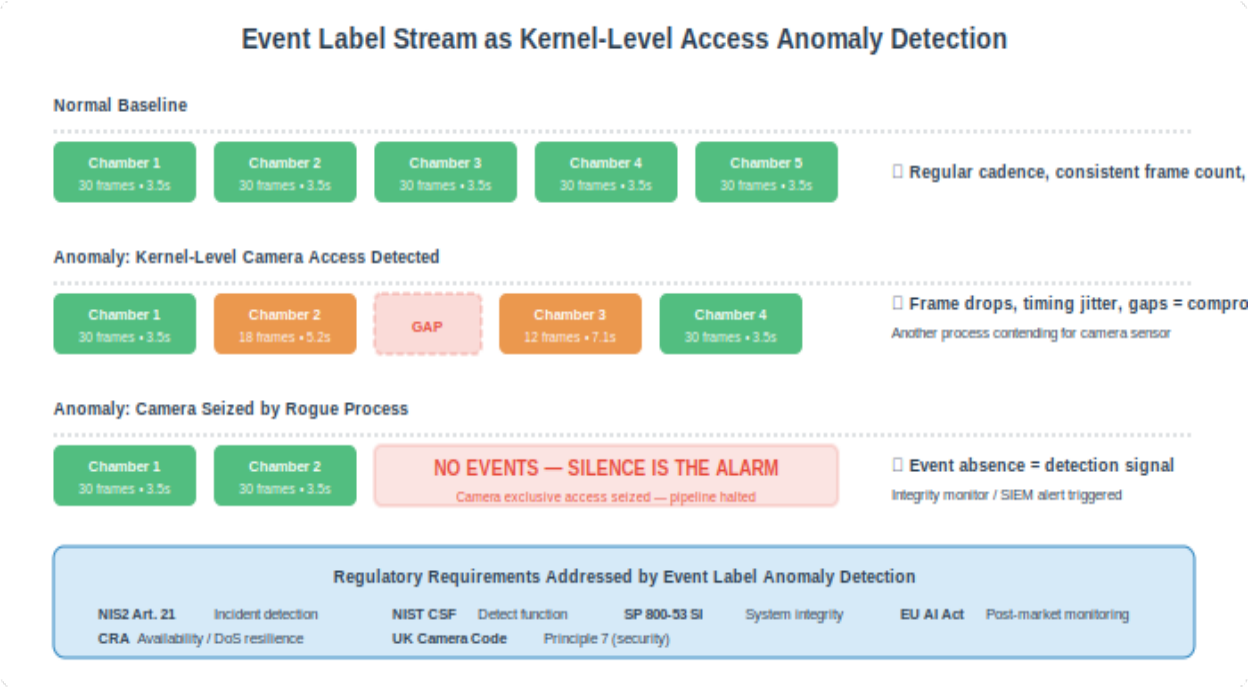
During normal operation, Chamber Sentinel produces a predictable stream of event labels: one event per chamber burn, at regular ~3.5-second intervals, with consistent frame counts (30 per chamber) and stable capture rates (~8.5 fps). This baseline is a tamper-evident signal. If a kernel-level attacker, rootkit, or rogue process accesses the camera outside of Chamber Sentinel's pipeline, the event label stream will exhibit detectable anomalies: **frame drops or gaps** (another process contending for the camera sensor produces backpressure or steals frames from the ImageReader queue); **timing irregularities** (chamber intervals stretching or compressing as CPU and ISP resources are shared with an unauthorised consumer); **missing chambers** (if the attacker seizes exclusive camera access, Chamber Sentinel's pipeline stops producing events entirely — an absence that is itself a detection signal); and **capture rate changes** (the effective fps dropping below the ~8.5 baseline as the ISP services multiple consumers).

This means the event label stream is not merely a record of what the camera detected — it is a continuous integrity attestation of the camera pipeline's health. A monitoring system (the planned M4 integrity monitor, or an external SIEM consuming the event stream) can compare the observed event cadence against the expected baseline and flag deviations as potential kernel-level compromise indicators.

From a regulatory standpoint, this property addresses a gap that no conventional camera system can fill. Regulations like NIS2 (Article 21 — incident detection), the NIST CSF Detect function, NIST SP 800-53 SI (System and Information Integrity) controls, and the EU AI Act's post-market monitoring requirements all demand mechanisms to detect compromise or degradation. Conventional cameras have no self-monitoring capability — they record until someone notices they've been tampered with. Chamber Sentinel's event label baseline provides a continuous, low-bandwidth, auditable signal that can be integrated into existing security monitoring infrastructure to satisfy these detection requirements.

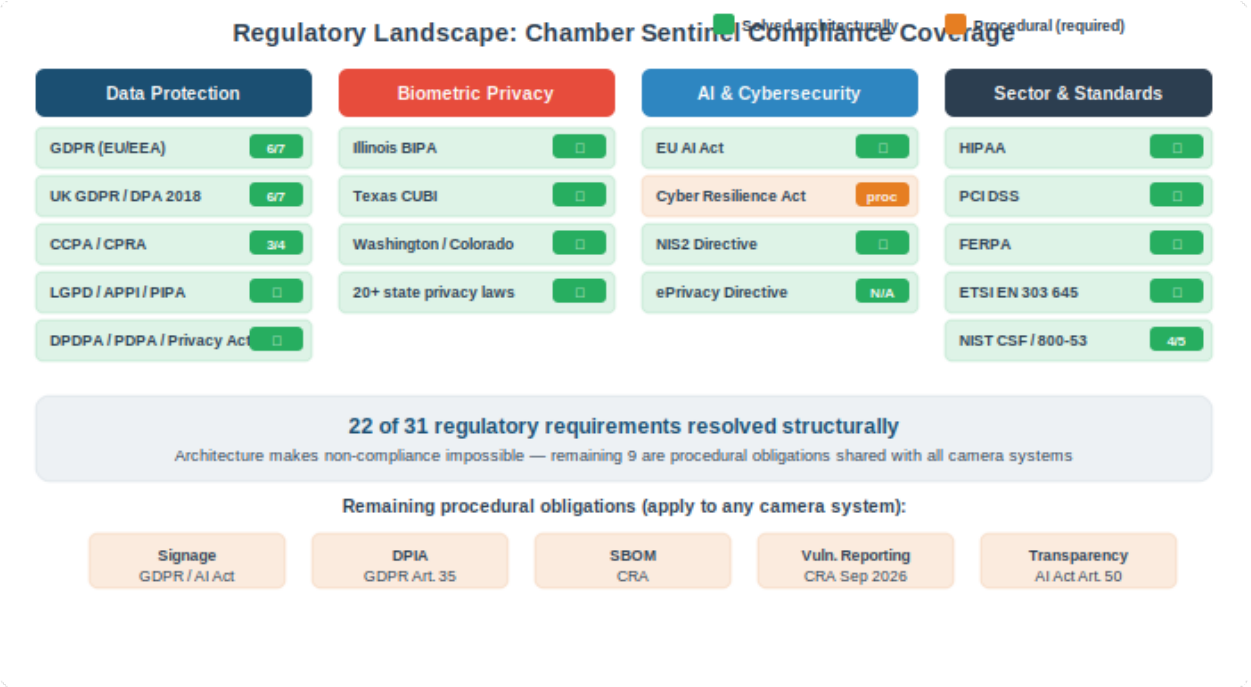
This also strengthens the system's position under the Cyber Resilience Act's requirement for products to "protect the availability of essential functions, including the resilience of and

protection against denial-of-service attacks" — the event stream makes denial-of-service (camera hijacking) detectable even when it cannot be prevented at the application layer.



9.16 Compliance Advantage Summary

The following table maps regulatory requirements to how Chamber Sentinel's architecture resolves them, compared to conventional camera systems.



Regulation	Requirement	Conventional Camera	Chamber Sentinel

		Problem	Resolution
GDPR Art. 5(1)(c)	Data minimisation	Stores weeks of footage beyond what is necessary	Footage destroyed in ~3.5s; output is non-personal event labels
GDPR Art. 5(1)(e)	Storage limitation	Must define, enforce, and audit retention policies	No retention to manage — destruction is cryptographic, not procedural
GDPR Art. 17	Right to erasure	Must locate and delete footage across storage systems	Automatically satisfied — footage destroyed before request possible
GDPR Art. 15	Right of access	Must review, redact, and provide footage	No footage exists to provide — controller confirms destruction
GDPR Art. 25	Privacy by design	Requires bolted-on controls over persistent data	Encryption, burn, no-network, schema constraints built in
GDPR Art. 33	Breach notification	Breach exposes weeks of footage — catastrophic impact	Breach exposes at most ~3.5s of encrypted frames
GDPR Ch. V	International transfers	Cloud cameras transfer data cross-border	No network — no transfer — no Chapter V obligations
EU AI Act Art. 5	Prohibited practices	Facial recognition and biometric ID face prohibition	Non-biometric object detection — outside prohibited category
EU AI Act	Penalty exposure	Up to €35M or 7% turnover for biometric AI	Architecture avoids high-risk and prohibited classifications
CRA	Secure by design	Network-connected cameras expand attack surface	No network, no IPC, encrypted memory, minimal surface
CRA	Breach impact	Persistent data means large breach scope	Ephemeral data limits breach to one chamber window
NIS2	Supply chain risk	Cloud cameras introduce third-party dependencies	No cloud, no network, no third-party data flow
CCPA/CPRA	Right to delete	Must process deletion requests across stored footage	Automatic — footage does not persist beyond ~3.5s
CCPA/CPRA	Sensitive PI exposure	Biometric data stored and potentially transmitted	Biometric data (if any) destroyed within seconds
BIPA	Retention/destruction	Most violations: retaining biometric data without schedule	Destruction is architectural — cannot retain even if desired

BIPA	Penalty exposure	Per-person statutory damages ($5,000 per violation)	Reduced exposure — no retention means fewer violation categories
Texas CUBI	Biometric penalties	$1B+ settlements for retention violations	Automatic destruction eliminates retention violations
US Wiretap Act	Audio consent	Audio-capable cameras trigger consent requirements	JPEG-only capture — no audio processing
Workplace laws	Function creep	Footage used beyond stated security purpose	No footage to misuse — structural safeguard
HIPAA	PHI protection	Camera footage as persistent PHI in clinical settings	PHI exposure limited to ~3.5s; event-only output
PCI DSS	Data encryption	Footage often unencrypted at rest	AES-256-GCM per-chamber encryption by default
FERPA	Education records	School footage creates education record obligations	No footage retained — no records created
Critical infra.	Network attack surface	IP cameras as network entry points	No network connectivity — cannot be pivot point
LGPD/APPI/ PIPA	International compliance	Each jurisdiction requires separate retention policies	Automatic destruction satisfies all retention requirements
ETSI EN 303 645	IoT baseline security	Connected devices require extensive hardening	Architecture inherently satisfies majority of requirements

Key finding: Across 31 regulatory requirements examined, Chamber Sentinel's burn-first architecture addresses 22 at the architectural level — meaning the architecture significantly reduces the compliance burden for these requirements compared to conventional camera systems. The remaining 9 (signage, DPIAs, transparency disclosures, consent mechanisms, SBOMs, vulnerability reporting) are procedural obligations that apply to any camera system regardless of architecture. The pattern across all frameworks examined is consistent: burn-first processing converts the compliance requirements most frequently violated in practice — retention management, access request handling, breach impact, cross-border transfers — from ongoing operational obligations into properties of the system's design. This does not eliminate the regulatory burden, but it concentrates the remaining compliance work on procedural requirements where the deployer's specific context (jurisdiction, sector, purpose) determines the obligation.

10. Limitations

Detection model accuracy is poor. The SSD MobileNet V1 quantized model produces confidence scores of 13–20% for correct detections. The model sometimes misclassifies persons as animals or unknown objects. This is a known limitation of INT8 quantized models with low confidence calibration. A production system requires a better model.

Integrity monitor not yet active. The M4 milestone (wiring the 5 monitors into a ForegroundService) has not been completed. The current prototype processes frames and burns chambers but does not detect compromise.

StrongBox not yet integrated. K_w is generated and stored in Rust process memory (mlock'd), not wrapped under a hardware key in StrongBox. M5 adds this.

GC residue is undersolved. The Kotlin-side byte array copy from ImageReader is not explicitly zeroed after JNI transfer. It persists until garbage collection, which is non-deterministic — under low memory pressure, the array could persist for seconds or longer. Calling Arrays.fill(frameBytes, 0) before dropping the reference is trivial and its absence weakens the security posture for no clear reason. This should be fixed immediately. The gralloc allocator may also reuse buffers without zeroing — this is an Android-level issue not addressable from user space.

Camera not truly headless. The Camera2 API requires an active Activity in the foreground. Background camera access is blocked by Android 14. If the screen locks, the camera stops. A production system would need a ForegroundService with camera type to maintain access.

Burn rate slower than designed. The target was 1 chamber per second. Measured rate is ~1 chamber per 3.5 seconds due to inference latency and GC pauses. The exposure window is therefore 3.5 seconds, not 1 second.

No on-device residue verification — this is the critical gap. The Rust substrate has 10 integration tests passing (including residue score = 0.0 after burn), but these run on the host machine, not on the A55. The entire security claim is "zero frames retained after burn," yet the actual device has residue sites that host tests cannot model: Android's dalvik heap (Kotlin byte arrays awaiting GC), the JPEG decoder's internal buffers, gralloc's buffer pool (may recycle without zeroing), the Exynos ISP's DMA ring buffers, and L1/L2 cache lines containing stale frame data. A post-burn memory dump analysis on the device (using adb shell dumpsys meminfo or a rooted device's /proc/pid/mem) is required to validate the claim. Until this is done, "zero residue" is a substrate-level claim, not a device-level claim.

Self-referential foundation. This paper builds on [1] by the same author.

Event labels reveal life patterns. "Person at door at 8 AM every weekday" is itself information. This is accepted residue.

11. Measured Results Summary

Note: *All residue and retention claims in this section are substrate-level measurements (Rust integration tests on host machine) unless otherwise stated. On-device verification on the Samsung A55 has not been performed. See Section 10 for the full list of unverified residue sites.*

Metric	Design target	Measured on A55
Frame rate	30 fps	30 fps (capture), ~8.5 fps (effective)
Frames per chamber	30	30
Chamber burn interval	1 second	~3.5 seconds
Frames dropped	0	0 (steady-state), 28–30 (warmup)
Chambers per 30s	30	~8–9

Inference time	< 50ms	~65ms
Model size	< 10MB	4.2MB
Detection confidence (person)	> 70%	13–14% (INT8 artifact)
APK size	—	~14MB (debug)
Rust .so size	—	630KB (stripped, release)
Forward secrecy	Per-chamber	Verified (unique K_w)
Post-burn residue (Rust tests)	0.0	0.0 (10 tests passing)
Network egress from app	0 bytes	0 bytes (no INTERNET)

12. Related Work

Puncturable encryption [11] provides formal constructions for forward-secret encryption where keys can be “punctured” to revoke decryption capability. Chamber Sentinel’s per-window key rotation is a coarse-grained instance.

TEE-based camera and sensor processing. CameraGuard [14] processes camera frames inside ARM TrustZone enclaves, providing hardware-grade isolation from the OS kernel — stronger than application-layer encryption. TrustOTP [13] demonstrates TEE-based secure processing on mobile devices for OTP generation (not camera-specific, but relevant for the TEE-on-mobile architecture pattern). The tradeoff between TEE-based and application-layer approaches is deployability versus isolation strength. TEE-based systems protect against root-level attackers (the K_w in process memory vulnerability identified in Section 2), but require hardware support, vendor cooperation, and platform-specific development. Chamber Sentinel runs on any Android device with the Camera2 API — no TEE, no vendor support, no special hardware. This is a deliberate design choice: broader deployability at the cost of weaker isolation against root-level attackers. A future iteration could combine both approaches — wrapping K_w in a StrongBox-backed key (M5 milestone) would provide hardware-grade key protection while retaining the burn-first architecture’s data-handling properties.

Haven [8] uses phone sensors as a physical intrusion detection system. It stores evidence rather than destroying it — a fundamentally different design point.

Covert channel analysis [12] provides formal methods for bounding information leakage through restricted APIs.

13. Conclusion

Chamber Sentinel demonstrates that the Chambers burn-first model [1] extends to continuous visual sensor data. The prototype running on a Samsung A55 confirms the core pipeline: camera frames are captured, encrypted, processed by an on-device detection model, and burned — every 3.5 seconds, with zero frames retained after burn (substrate-level; on-device verification pending).

The implementation reveals gaps between design and reality: inference is slower than budgeted (65ms vs 50ms target), confidence calibration is poor (13–14% for correct detections), the burn interval is 3.5× longer than designed (3.5s vs 1s), and several hardening measures (StrongBox, integrity monitor) are not yet active.

The traffic baseline insight remains the paper’s strongest contribution: removing all legitimate network traffic makes exfiltration trivially detectable for application-level threats. This is a design

principle, not an implementation detail, and it applies to any sensor processing system where the output is structurally smaller than the input.

The fundamental question — is event-label-only output sufficient for real-world monitoring? — cannot be answered by this paper. It depends on model accuracy (the prototype uses a deliberately lightweight model to validate the pipeline, not for production deployment), user expectations (untested), and deployment context. What the prototype demonstrates is the feasibility and limits of burn-first camera processing on commodity Android hardware: the camera sees, understands, and forgets. Whether forgetting is acceptable is a product question, not an engineering one.

References

[1] A. Ganguli, "Chambers: Sealed Ephemeral Worlds for Private Cognition," 2026. https://github.com/therealgulkorinaga/chamber/blob/main/Chambers%20Position%20Paper.pdf

[2] Signal Protocol. Double Ratchet Algorithm.

[3] Android Camera2 API.

[4] Samsung Knox Security.

[5] Android Keystore System.

[6] ARM Cryptographic Extensions.

[7] Video4Linux2 API.

[8] Guardian Project, Haven.

[9] NIST SP 800-193, Platform Firmware Resiliency Guidelines.

[10] Wei et al. "Reliably Erasing Data From Flash-Based Solid State Drives." FAST 2011.

[11] Green and Miers. "Forward Secure Asynchronous Messaging from Puncturable Encryption." IEEE S&P 2015.

[12] Lampson. "A Note on the Confinement Problem." CACM 1973.

[13] Sun et al. "TrustOTP: Transforming Smartphones into Secure One-Time Password Tokens." ACM CCS 2015.

[14] Li et al. "CameraGuard: Securing Cameras in the Era of IoT." IEEE IoTJ 2020.

[15] TensorFlow Lite.

[16] SSD MobileNet V1 COCO.